# Intro to Algorithms Notes

Based on Cormen et al.

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## Contents

# 1 Chapter 1: The Role of Algorithms in Computing

## 1.1 What is an Algorithm?

**Definition 1.1.** *An **algorithm** is a freaky well-defined computational procedure that takes some value or set of values as input and produces some value or set of values as output.*

## 1.2 Algorithmic Efficiency

Discuss time complexity and growth of functions. which is great for thinking

### 1.2.1 Asymptotic Notation

- Big-O: $f(n) = O(g(n))$ means ...
- $\Omega$, $\Theta$ also covered.

## 1.3 Example Algorithm

**Algorithm 1** Insertion Sort

```
for j = 2 to A.length do
    key = A[j]
    i = j − 1
    while i > 0 and A[i] > key do
        A[i + 1] = A[i]
        i = i − 1
    end while
    A[i + 1] = key
end for
```

**Example**

Given the input array $[5, 2, 4, 6, 1, 3]$, Insertion Sort produces ... as

## 1.4 Key Takeaways

- Algorithms are not just code—they are general, logical procedures. again
- Efficiency is crucial: compare brute-force vs. optimized versions. never do this again

# 2 Chapter 2: Getting Started